Name: _____ Date: _____

1. Find $f'(x)$ if $f(x) = e^{1-x^3}$.

2. Find $f'(x)$ if $f(x) = \dfrac{e^x}{x^2+1}$.

3. Find $f'(x)$ if $f(x) = e^{1/x} + \dfrac{1}{e^x}$.

4. Find $f'(x)$ if $f(x) = e^{-2x} \ln x$.

5. Find $f'(x)$ if $f(x) = e^{3x} \tan \sqrt{x}$.

6. Find $f'(x)$ if $f(x) = \sec^2(e^{-4x})$.

7. Find $f'(x)$ if $f(x) = \ln(\csc(e^{3x}))$.

8. Use implicit differentiation to find $y\prime$ if $xe^y + 2x - \ln(y+1) = 3$.

9. Find an equation of the tangent line to $y = x - e^{-x}$ that is parallel to $6x - 2y = 7$.

10. For $f(x) = xe^{-x}$, find the local extrema, intervals of increase and decrease, concavity, and inflection points. Sketch the graph.

11. The rate R at which a tumor grows is related to its size x by $R = rx\ln(K/x)$, where $r, K > 0$. Show that growth is most rapid when $x = e^{-1}K$.

12. A nerve fiber has axon radius r and insulation radius R, with $0 < r < R$. Its impulse velocity is $v = -k(r/R)^2\ln(r/R)$, where $k > 0$. Find the ratio r/R that maximizes v.

13. Evaluate (a) $\int \frac{1}{4-5x}\,dx$; (b) $\int_{-1}^{0} \frac{1}{4-5x}\,dx$.

14. Evaluate (a) $\int \frac{4x}{x^2-9}\,dx$; (b) $\int_{1}^{2} \frac{4x}{x^2-9}\,dx$.

15. Evaluate (a) $\int e^{-4x}\,dx$; (b) $\int_1^3 e^{-4x}\,dx$.

16. Evaluate (a) $\int \cot(x/3)\,dx$; (b) $\int_{3\pi/2}^{9\pi/4} \cot(x/3)\,dx$.

17. Evaluate $\int (\tan 3x + \sec 3x)\, dx$.

18. Find the area bounded by $y = e^{2x}$, $y = 0$, $x = 0$, and $x = \ln 3$.

19. The region bounded by $y = \sec x$, $x = -\pi/3$, $x = \pi/3$, and $y = 0$ is revolved about the x-axis. Find the volume.

20. A probability density function on $[a, b]$ is nonnegative and has integral 1. Determine c so that $f(x) = cx/(x^2 + 4)$ on $0 \leq x \leq 3$ is a probability density function.

21. In a circuit, the current is $I(t) = 10e^{-4t}$ amperes. If Q is charge in coulombs, $I = dQ/dt$. (a) Find $Q(t)$ if $Q(0) = 0$. (b) Find the charge after a long period of time.

22. Find $f'(x)$ if $f(x) = 7^x$.

23. Find $f'(x)$ if $f(x) = \log_3 |6x - 7|$.

24. Find $f'(x)$ if $f(x) = \log \sqrt{x^2 + 1}$.

25. Find $f'(x)$ if $f(x) = x^{\pi}\pi^{x}$.

26. Find $f'(x)$ if $f(x) = 4^{\sec 3x}$.

27. Evaluate $\int 10^{3x}\,dx$.

28. Evaluate $\int \dfrac{2^x}{2^x+1}\,dx$.

29. Evaluate $\displaystyle\int \frac{1}{x \log x}\, dx$.

30. Evaluate (a) $\displaystyle\int \pi^x\, dx$; (b) $\displaystyle\int x^4\, dx$; (c) $\displaystyle\int x^\pi\, dx$; (d) $\displaystyle\int \pi x\, dx$.

31. Find the area bounded by $y = 2^x$, $x + y = 1$, and $x = 1$.

32. The region under $y = 3^{-x}$ from $x = 1$ to $x = 2$ is revolved about the x-axis. Find the volume.

33. A city has population 500,000. Its population increases at 5% per year, with rate proportional to population. What will the population be in 10 years?

34. Assume one quarter acre is required to feed one person and 10 billion acres are tillable, giving a maximum sustainable population of 40 billion. At the beginning of 1980 the population was 4.5 billion. If population grows continuously at 2% per year, with rate proportional to population, when is the maximum reached?

35. A metal plate cools from 180°F to 150°F in 20 minutes in air at 60°F. Use Newton's law of cooling to estimate (a) its temperature after one hour and (b) when it reaches 100°F.

36. At 600 K, nitrogen dioxide decomposes according to $dy/dt = -0.05y^2$, where $y(t)$ is concentration in moles per liter and t is in seconds. Express y in terms of t and its initial concentration $y_0 > 0$.

37. Let $I_0 > 0$ be the incoming radiation intensity, $\rho(h)$ the atmospheric density, and $k > 0$ an absorption constant. Define $f(x) = k\int_0^x \rho(h)\,dh$ and $I(x) = I_0 e^{-f(x)}$. Show that $dI/dx = -k\rho(x)I$.

Tommy's Questions

Tommy 1. Let $f(x) = xe^{-x}$ for $x \geq 0$.

(a) Find $f\prime(x)$ and $f\prime\prime(x)$.

(b) Find the absolute maximum value of f on $[0, \infty)$.

(c) Find the exact area under $f\prime$ and above the x-axis on $[0, 1]$.

(d) Find $b > 1$ such that $\int_0^b f\prime(x)\,dx = \int_b^\infty f\prime(x)\,dx$, or show that no such b exists. Interpret the latter integral as a limit.

Tommy 2. An object cools according to Newton's law in a room of constant, unknown temperature. Its temperature is 100°C initially, 70°C after 10 minutes, and 55°C after 20 minutes. How many minutes after cooling begins does its temperature first reach 41.875°C?

Tommy 3. For $1 \le x \le \sqrt{2}$, let $T(x)$ be the value of the convergent infinite power tower

$$T(x) = x^{x^{x^{\cdot^{\cdot^{\cdot}}}}}.$$

Use the continuous branch with $T(1) = 1$ and $T(\sqrt{2}) = 2$. Evaluate exactly

$$\lim_{b \to \sqrt{2}^-} \int_1^b \frac{T(x)}{x[1 - T(x) \ln x]}\, dx.$$

Answer Key

1. $-3x^2e^{1-x^3}$.

2. $-\frac{2xe^x}{(x^2+1)^2}+\frac{e^x}{x^2+1}$.

3. $-e^{-x}-\frac{e^{\frac{1}{x}}}{x^2}$.

4. $-2e^{-2x}\log(x)+\frac{e^{-2x}}{x}$.

5. $3e^{3x}\tan(\sqrt{x})+\frac{\left(\tan^2\left(\sqrt{x}\right)+1\right)e^{3x}}{2\sqrt{x}}$.

6. $-8e^{-4x}\tan\left(e^{-4x}\right)\sec^2\left(e^{-4x}\right)$.

7. $-3e^{3x}\cot\left(e^{3x}\right)$.

8. $y\prime=-\frac{(y+1)(e^y+2)}{x(y+1)e^y-1}$.

9. $y=3x+2\ln 2-2$, at $(-\ln 2,-\ln 2-2)$.

10. Local maximum $(1,e^{-1})$; no local minimum. Increasing on $(-\infty,1)$, decreasing on $(1,\infty)$. Concave down on $(-\infty,2)$ and up on $(2,\infty)$; inflection $(2,2e^{-2})$. Passes through $(0,0)$; $f(x)\to-\infty$ as $x\to-\infty$ and $f(x)\to 0^+$ as $x\to\infty$.

11. $R\prime=r[\ln(K/x)-1]$ vanishes only at $x=K/e$; $R\prime\prime=-r/x<0$. Maximum rate rK/e.

12. $r/R=e^{-1/2}\approx 0.606531$; $v_{\max}=k/(2e)$.

13. (a) $-\frac{1}{5}\ln|4-5x|+C$; (b) $\frac{1}{5}\ln(9/4)$.

14. (a) $2\ln|x^2-9|+C$; (b) $2\ln(5/8)$.

15. (a) $-\frac{e^{-4x}}{4}+C$; (b) $-\frac{1-e^8}{4e^{12}}$.

16. (a) $3\ln|\sin(x/3)|+C$; (b) $-\frac{3}{2}\ln 2$.

17. $-\frac{1}{3}\ln|\cos 3x|+\frac{1}{3}\ln|\sec 3x+\tan 3x|+C$.

18. 4.

19. $2\pi\sqrt{3}$.

20. $c=2/\ln(13/4)$.

21. (a) $Q(t)=\frac{5}{2}(1-e^{-4t})$ coulombs; (b) $\frac{5}{2}$ coulombs.

22. $7^x\log(7)$.

23. $\frac{6}{(6x-7)\log(3)}$.

24. $\frac{x}{(x^2+1)\log(10)}$.

25. $\pi^x x^\pi \log(\pi) + \frac{\pi \pi^x x^\pi}{x}$.

26. $3 \cdot 4^{\sec(3x)} \log(4) \tan(3x) \sec(3x)$.

27. $10^{3x}/(3 \ln 10) + C$.

28. $\ln(2^x + 1)/\ln 2 + C$.

29. $(\ln 10) \ln|\ln x| + C$.

30. (a) $\pi^x/\ln \pi + C$; (b) $x^5/5 + C$; (c) $x^{\pi+1}/(\pi + 1) + C$; (d) $\pi x^2/2 + C$.

31. $1/\ln 2 - 1/2$.

32. $4\pi/(81 \ln 3)$.

33. $500000e^{0.5} \approx 824361$ people.

34. $50 \ln(40/4.5) \approx 109.240$ years after the beginning of 1980, during 2089.

35. (a) $60 + 120(3/4)^3 = 110.625°$F; (b) $20 \ln(1/3)/\ln(3/4) \approx 76.377$ minutes after cooling begins.

36. $y(t) = y_0/(1 + 0.05 y_0 t)$ for $t \geq 0$.

37. $I\prime(x) = -I_0 e^{-f(x)} f\prime(x) = -k\rho(x)I(x)$.

Tommy's Questions: Solutions

Tommy 1. (a) By the product rule,

$$f\prime(x) = e^{-x}(1-x), \qquad f\prime\prime(x) = e^{-x}(x-2).$$

(b) The first derivative is positive for $0 \le x < 1$ and negative for $x > 1$. Thus the absolute maximum is $f(1) = 1/e$.
(c) Since $f\prime \ge 0$ on $[0,1]$, the requested area is

$$\int_0^1 f\prime(x)\,dx = f(1) - f(0) = 1/e.$$

(d) The two signed integrals are

$$\int_0^b f\prime(x)\,dx = be^{-b}, \qquad \lim_{M\to\infty}\int_b^M f\prime(x)\,dx = -be^{-b}.$$

For every finite $b > 1$, the first is positive and the second negative, so no such b exists. Equality of areas would use the negative of the second integral; it is not equality of these signed integrals.

Tommy 2. Write θ for room temperature and $q = e^{10k}$, where $k < 0$. Newton's law gives

$$100 - \theta = A, \quad 70 - \theta = Aq, \quad 55 - \theta = Aq^2.$$

Subtract successive readings:

$$30 = A(1-q), \qquad 15 = Aq(1-q).$$

Hence $q = 1/2$, $A = 60$, and $\theta = 40$. Therefore

$$T(t) = 40 + 60\,2^{-t/10}.$$

The target temperature satisfies

$$41.875 - 40 = 1.875 = 60/32 = 60\,2^{-5}.$$

Thus $t/10 = 5$ and $\boxed{t = 50 \text{ minutes}}$.

Tommy 3. The tail of the tower is the same tower, so $T = x^T$. Taking natural logarithms,

$$\ln T = T \ln x.$$

Implicit differentiation for $1 < x < \sqrt{2}$ gives

$$\frac{T\prime}{T} = T\prime \ln x + \frac{T}{x}, \qquad \frac{T\prime}{T} = \frac{T}{x(1 - T\ln x)}.$$

The integrand is therefore the derivative of $\ln T$. For $b < \sqrt{2}$,

$$\int_1^b \frac{T(x)}{x[1 - T(x)\ln x]}\,dx = \ln T(b) - \ln T(1).$$

By continuity of the specified branch, the limit is $\boxed{\ln 2}$. The denominator at the upper endpoint is $1 - \ln 2 > 0$, so the expression also extends continuously to that endpoint.

Source List

These questions come from Sections 7.3, 7.4, 7.5, and 7.6. Try the surrounding problems for more practice.

Guide question	Textbook source
1	7.3 #4
2	7.3 #11
3	7.3 #15
4	7.3 #19
5	7.3 #25
6	7.3 #27
7	7.3 #30
8	7.3 #32
9	7.3 #36
10	7.3 #40
11	7.3 #53
12	7.3 #57
13	7.4 #2
14	7.4 #3
15	7.4 #5
16	7.4 #8
17	7.4 #34
18	7.4 #37
19	7.4 #40
20	7.4 #45
21	7.4 #51
22	7.5 #1
23	7.5 #6
24	7.5 #12
25	7.5 #18
26	7.5 #22
27	7.5 #31
28	7.5 #35
29	7.5 #37
30	7.5 #41
31	7.5 #43
32	7.5 #44
33	7.6 #4
34	7.6 #5
35	7.6 #6
36	7.6 #18
37	7.6 #21
Tommy 1	Original; not in the textbook
Tommy 2	Original; not in the textbook
Tommy 3	Original; not in the textbook